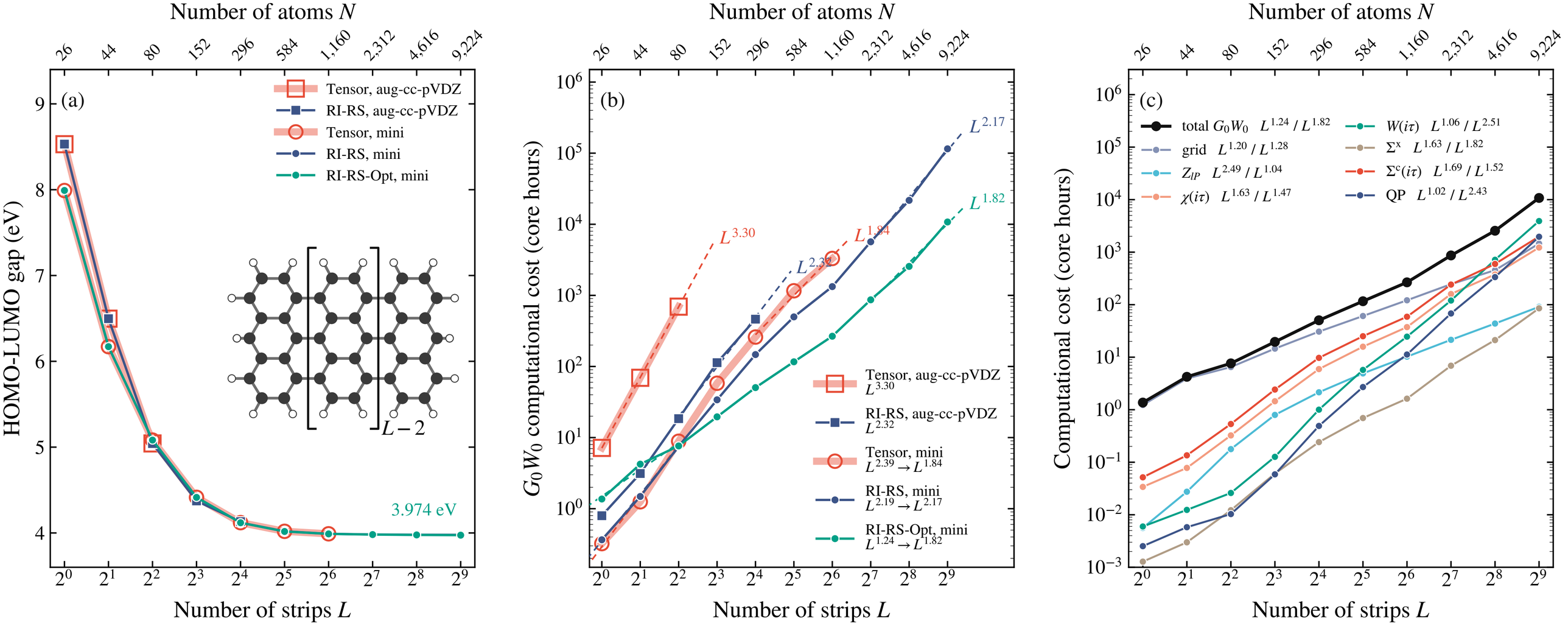
Number of atoms N
26
44
80
152
296
584
1,160
2,312
4,616
9,224
(a)
Tensor, aug-cc-pVDZ
RI-RS, aug-cc-pVDZ
Tensor, mini
RI-RS, mini
RI-RS-Opt, mini
HOMO-LUMO gap (eV)
9
8
7
6
5
4
3.974 eV
L − 2
Number of strips L
(b)
G_0W_0 computational cost (core hours)
$L^{3.30}$
$L^{2.32}$
$L^{1.84}$
$L^{2.17}$
$L^{1.82}$
Tensor, aug-cc-pVDZ
$L^{3.30}$
RI-RS, aug-cc-pVDZ
$L^{2.32}$
Tensor, mini
$L^{2.39} \rightarrow L^{1.84}$
RI-RS, mini
$L^{2.19} \rightarrow L^{2.17}$
RI-RS-Opt, mini
$L^{1.24} \rightarrow L^{1.82}$
(c)
Computational cost (core hours)
total G_0W_0 $L^{1.24}$ / $L^{1.82}$
grid $L^{1.20}$ / $L^{1.28}$
Z_{lP} $L^{2.49}$ / $L^{1.04}$
$\chi(i\tau)$ $L^{1.63}$ / $L^{1.47}$
$W(i\tau)$ $L^{1.06}$ / $L^{2.51}$
Σ^x $L^{1.63}$ / $L^{1.82}$
$\Sigma^c(i\tau)$ $L^{1.69}$ / $L^{1.52}$
QP $L^{1.02}$ / $L^{2.43}$